2 Car harness and box edge

Everything outside the box: battery + F1, key, pedal,
shifter, 5 motors, E-stop, sensors, dash, speakers.
Box plugs J1-J12 (J1-J8 mate the stock plugs, TBD #17).

File: harness.kicad_sch

3 Power and safety chain

K1 contactor and its coil chain: key, E-stop (2 NC),
Q1 hold MOSFET. F2 key feed, fuse block F3-F8,
ground star TB1, motor bus TB2, battery divider.

File: power.kicad_sch

4 Low-voltage rails

U1 LM2596 5V_LOGIC, U2 AMS1117 3V3_SENSE,
U3 5V_ACC and U4 12V_ACC for accessories.

File: rails.kicad_sch

5 Drive, variant A (4 + 1 IBT-2)

U10-U13 IBT-2 per wheel, U14 steering IBT-2.
470 µF each, enable pull-down per board,
IS current-sense networks to the ADS1115s.

File: drive.kicad_sch

6 Controller and inputs

ESP32 (nets from include/pins.h), ELRS receiver,
pedal input protection, wheel-speed pull-ups,
DS18B20, DFPlayer serial.

File: controller.kicad_sch

7 I2C bus and switch inputs

2 x ADS1115 current sense, PCF8574 switches
(shifter, horn, E-stop sense, seat), MPU-6050,
PCA9685 accessory PWM.

File: i2c_io.kicad_sch

8 Accessories: fans, lights, sound

MOSFET board for lights and fans, 12 V branch
fuses F9/F10, music amp, DFPlayer + alert speaker.

File: accessories.kicad_sch

How to read it

Every pin has a stub and a net label: the label names the net, so each pin says where it goes. Labels with a flag shape are global (used on more than one sheet); plain text labels stay on one sheet. Each box above is a sheet: double-click it in KiCad to open it (or page down in the PDF). Location field: box, A1 (the interface board), car, dash. Source field: owned, ordered 24 Oct, stock car, to buy, TBD. Tables: NETS.md, CONNECTORS.md, WIRE_LIST.md, BOM.md (same data).

Three safety layers (POWER_CHAIN.md)

1. A power chain a person can always cut, no software:
   F1, key SW1 and E-stop S2 (two NC blocks) in the K1 coil.
2. Drivers that default off: DRIVE_EN / STEER_EN pulled down
   at every board; K1 coil MOSFET Q1 pulled down.
3. The ESP32 only adds control on top, and can only open K1.

Open items (TBD)

#17 stock plug types and pin use: J1-J8 (car-facing plugs)
#33 motor pairing in the harness: J6/J7 splitter or not
#13 battery voltage and chemistry: D2 TVS value, BTS7960 27.5 V limit
#14 motor labels: per-motor current, variant A or B (#32)
#18 box size: variant A (5 boards) or B (2 boards)
#16 pedal type: switch, pot or 5 V hall (divider)
#15 IBT-2 IS resistor: config IS_RESISTOR_OHMS
Key switch rating (SW1 carries ~8 A worst case)
Shifter: signal-only or motor current (only signal can go to U24)
K1 precharge resistor: after the bench test
AS5600 cable length: I2C or analog OUT on U23 A1
Variant B current sensors (#32)

Revisions

0.3 2026-10-24 First KiCad version, from draft v0.2 (SVG).
Net names, connector/pin tables, wire list, BOM
matched to the 24 Oct order. Changes are listed
in README.md, 'Changes from draft v0.2'.

Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

Sheet: /
File: rideonesp32.kicad_sch

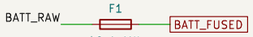
Title: RideOnESP32 v3 wiring schematic (index)

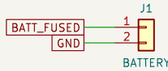
Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 1/8

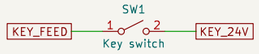
Left: what stays in the car (stock parts and our off-box parts). Right of each plug: into the box.
Plugs facing the stock harness (J1–J8) are TBD until the stock plug photos (#17).
J9–J12 are our own sealed plugs.

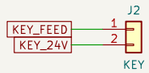
Battery and key



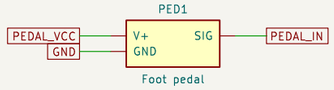


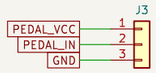






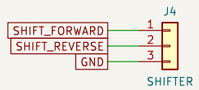
Pedal and shifter



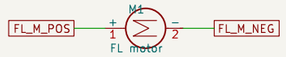








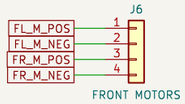
Front motors



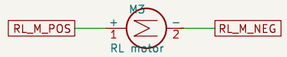






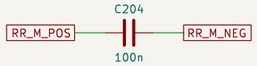


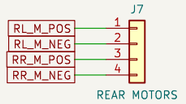
Rear motors



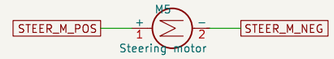




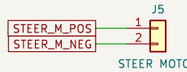


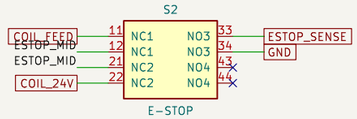


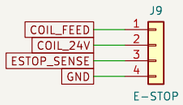
Steering motor, E-stop



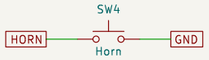




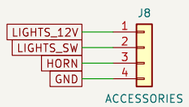




Horn and lights





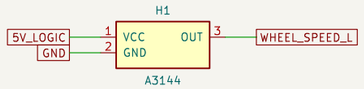


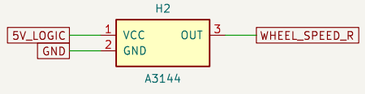
Going back to stock

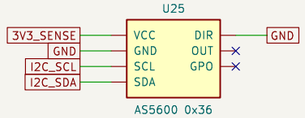
1. Key off. Unplug J1–J8 from the box.
2. Plug the stock HY2005RX controller into the same plugs (motor splitters: unplug, rejoin).
3. Unplug J9–J12; the stock car has none of them.

No stock wire is ever cut (#33).

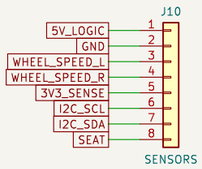
Sensors



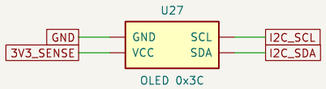


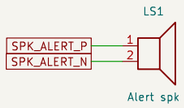


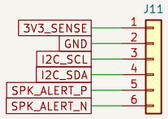




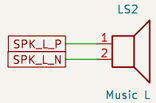
Dash

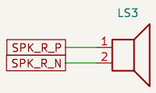


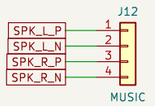




Music







Draft v0.3, not for building the car-facing side until the TBDs close

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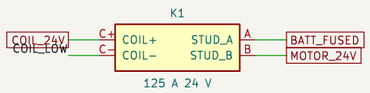
Sheet: /2 Car harness and box edge/
File: harness.kicad_sch

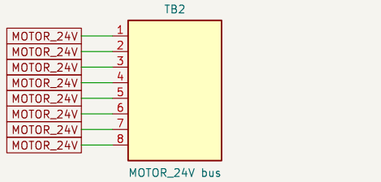
Title: Car harness and box edge

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 2/8

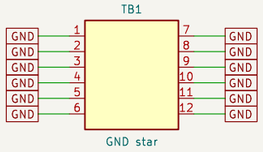
Layer 1, no software: key and E-stop in series with the K1 coil; either one opens K1 whatever the ESP32 does.
Q1 lets the ESP32 open K1 but never close it. Reset, crash or brown-out opens it (R2 pull-down).
Rails come from the key side, before K1, so the ESP32 stays up after an E-stop and can report why.

Main path

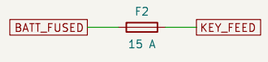


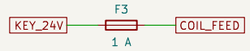




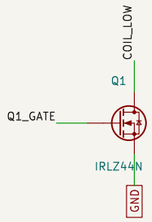


Coil chain









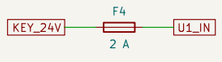


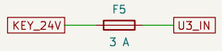


Coil chain

KEY_24V -> F3 1 A -> J9.1 -> S2 NC1 -> S2 NC2
-> J9.2 -> K1 coil -> Q1 -> GND.
Unplugging J9 opens K1 (fail-safe).
Q1 gate: GPIO12 via R1, R2 pull-down.

Fuse block (key side)

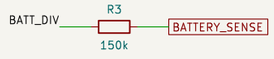
















Notes

Main current path

BT1 + -> F1 40 A -> J1 -> K1 stud A -> K1 stud B
-> TB2 -> B+ of U10-U14 (10 AWG).
B- of U10-U14 -> TB1 star -> J1 -> BT1 -.
The blade fuse block carries only the small
branches; it is not in the 40 A path.
K1 precharge: decide after the bench test
(5 x 470 µF behind a 125 A contactor).

Fuse rule

Every fuse protects the wire after it:
the wire is rated above the fuse.
F1 40 A: 10 AWG. F2 15 A: 14 AWG.
F6 10 A: 16 AWG. F3-F5, F7, F8: 20 AWG
or heavier. See WIRE_LIST.md.











Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

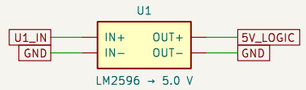
Sheet: /3 Power and safety chain/
File: power.kicad_sch

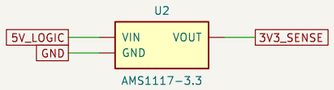
Title: Power and safety chain

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 3/8

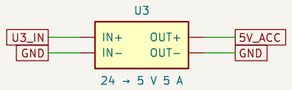
Logic and accessories never share a converter. 3V3_SENSE comes from 5V_LOGIC, not the ESP32's own 3.3 V pin.
Set each buck's output with a meter before connecting any load.

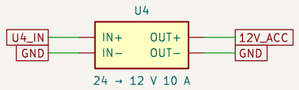
Logic





Accessories





Budget

Rail budget

5V_LOGIC U1 LM2596 ESP32, RX, halls, U2 ~0.5 A F4 2 A
3V3_SENSE U2 AMS1117 IBT-2 logic, I2C ~0.1 A (via F4)
5V_ACC U3 5 A DFPlayer, USB, LEDs ≤5 A F5 3 A
12V_ACC U4 10-20 A lights F9, amp F10 ≤10 A F6 10 A
Fuses are on the 24 V input side: 5 A at 5 V is about 1.2 A at 24 V.

Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

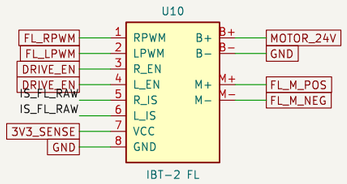
Sheet: /4 Low-voltage rails/
File: rails.kicad_sch

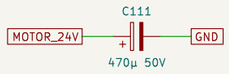
Title: Low-voltage rails

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 4/8

Layer 2: DRIVE_EN and STEER_EN are pulled down at every board, so boot, reset, flashing and crashes leave the drivers off.
IS sense: R_IS + L_IS joined (only one side conducts at a time) -> 4.7k -> ADS1115, 10 nF filter, clamp to 3.3 V.
Variant B (2 x MDD20A) replaces this sheet: see variant_b/.

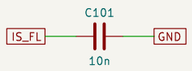
FL (front-left)





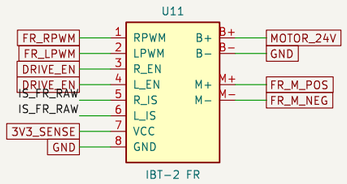


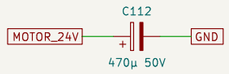






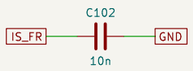
FR (front-right)





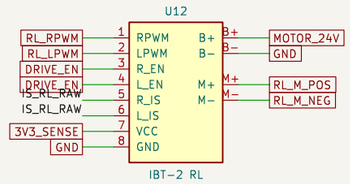


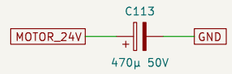




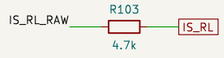


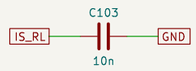
RL (rear-left)





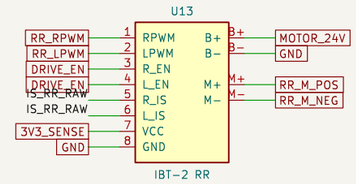


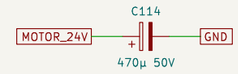






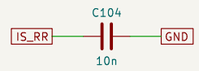
RR (rear-right)





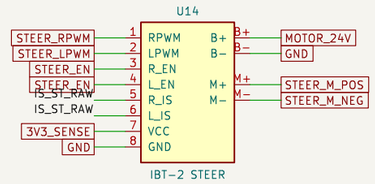


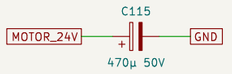






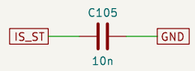
STEER (steering)













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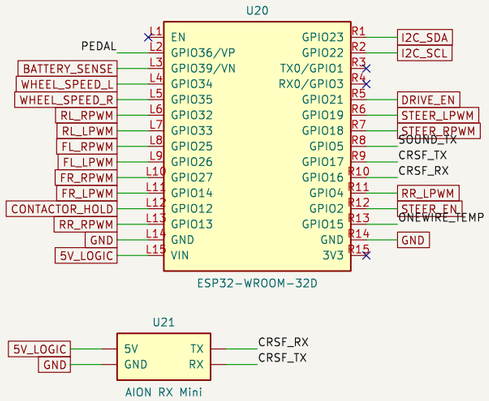
Sheet: /5 Drive, variant A (4 + 1 IBT-2)/
File: drive.kicad_sch

Title: Drive, variant A (4 + 1 IBT-2)

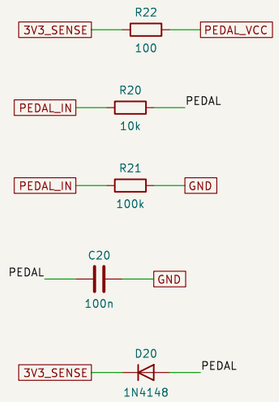
Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 5/8

Pin nets come from include/pins.h; CI keeps pins.h, PINOUT.md and WIRING.md in step and check_netlist.py checks this sheet.
Analog inputs on ADC1 only. GPIO36/39 carry analog only (they glitch when the ADC runs). Kill lines on pins quiet at boot.

ESP32



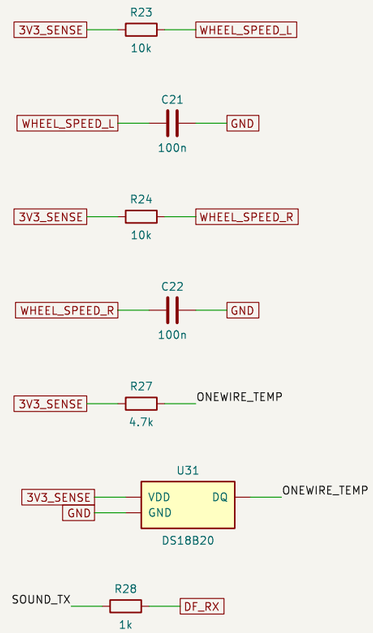
Pedal



Pedal input

Switch pedal: between V+ and SIG.
Pot: ends on V+ / GND, wiper on SIG.
5 V hall throttle: add a divider (TBD #16).
Shorted high reads as full pedal: open bench check (CLAUDE.md).

Wheel speed, temperature, sound



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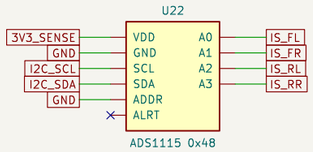
Sheet: /6 Controller and inputs/
File: controller.kicad_sch

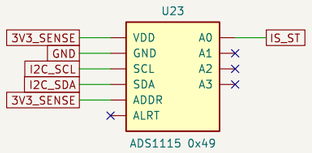
Title: Controller and inputs

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 6/8

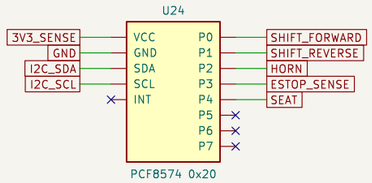
I2C on GPIO22 (SCL) / GPIO23 (SDA) at 3.3 V. Addresses: 0x20 0x36 0x3C 0x40 (+0x70 all-call) 0x48 0x49 0x68.
Off-box I2C (AS5600 on the steering column, OLED on the dash) is the weak spot: keep runs short, twisted with GND (#36).

Current sense ADCs





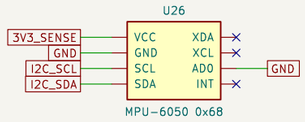
Switches

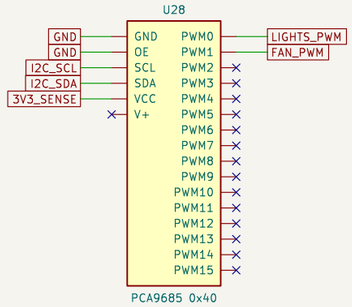


PCF8574 inputs

P0 shift forward, P1 shift reverse, P2 horn,
P3 E-stop sense (NO contact: 0 = pressed),
P4 seat switch. P5–P7 spare.
The E-stop itself cuts K1 in hardware;
P3 only tells the firmware why.

IMU, PWM, pull-ups







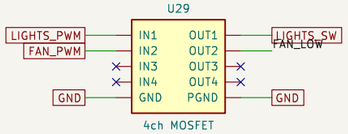


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Sheet: /7 I2C bus and switch inputs/		
File: i2c_io.kicad_sch		
Title: I2C bus and switch inputs		
Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 7/8

Phase 7. Nothing here is in the 24 Oct order except the inline fuse holders.
Music has its own amp with Bluetooth on that board only; the ESP32 never runs Wi-Fi or Bluetooth.

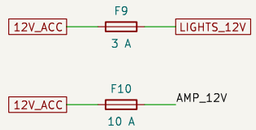
Switched loads

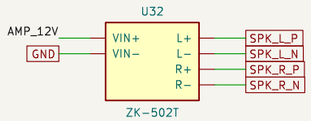






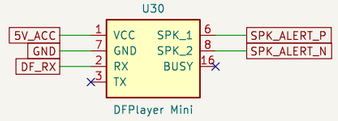
12 V branches







Car sounds



Audio

DFPlayer on 5V_ACC so speaker current stays
off the ESP32 supply. See docs/hardware/AUDIO.md.

Draft v0.3, not for building the car-facing side until the TBDs close

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Sheet: /8 Accessories: fans, lights, sound/
File: accessories.kicad_sch

Title: Accessories: fans, lights, sound

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 8/8